Quantification --- Forming Propositions from Predicates

Subjects to be Learned

- universe
- universal quantifier
- existential quantifier
- free variable
- bound variable
- scope of quantifier
- order of quantifiers

Contents

A predicate with variables is not a proposition. For example, the statement $x > 1$ with variable x over the universe of real numbers is neither true nor false since we don't know what x is. It can be true or false depending on the value of x.
For $x > 1$ to be a proposition either we substitute a specific number for x or change it to something like "There is a number x for which $x > 1$ holds", or "For every number x, $x > 1$ holds".

More generally, a predicate with variables (called an **atomic formula**) can be made a **proposition** by applying one of the following two operations to each of its variables:

1. assign a value to the variable
2. quantify the variable using a **quantifier** (see below).

For example, $x > 1$ becomes $3 > 1$ if 3 is assigned to x, and it becomes a true statement, hence a proposition.

In general, a quantification is performed on formulas of predicate logic (called wff), such as $\boldsymbol{x > 1}$ or $\boldsymbol{P(x)}$, by using quantifiers on variables. There are two types of quantifiers: universal quantifier and existential quantifier.

The **universal quantifier** turns, for example, the statement $x > 1$ to "for every object x in the universe, $x > 1$", which is expressed as "$\forall x\ x > 1$". This new statement is true or false in the universe of discourse. Hence it is a proposition once the universe is specified.

Similarly the **existential quantifier** turns, for example, the statement $x > 1$ to "for some object x in the universe, $x > 1$", which is expressed as "$\exists\ x\ x > 1$." Again, it is true or false in the universe of discourse, and hence it is a proposition once the universe is specified.

Universe of Discourse

The universe of discourse, also called **universe**, is the set of objects of interest. The propositions in the predicate logic are statements on objects of a universe. The universe is thus the domain of the (individual) variables. It can be the set of real numbers, the set of integers, the set of all cars on a parking lot, the set of all students in a classroom etc. The

universe is often left implicit in practice. But it should be obvious from the context.

The Universal Quantifier

The expression: $\forall x\, P(x)$, denotes the **universal quantification** of the atomic formula P(x). Translated into the English language, the expression is understood as: "*For all x, P(x)* holds", "*for each x, P(x)* holds" or "*for every x, P(x)* holds". $\forall$ is called the **universal quantifier**, and $\forall x$ means all the objects x in the universe. If this is followed by *P(x)* then the meaning is that *P(x)* is true for every object *x* in the universe. For example, "All cars have wheels" could be transformed into the propositional form, $\forall x\, P(x)$, where:

- *P(x)* is the predicate denoting: ***x* has wheels**, and
- the universe of discourse is only populated by cars.

Universal Quantifier and **Connective AND**
If all the elements in the universe of discourse can be listed then the universal quantification $\forall x\, P(x)$ is equivalent to the conjunction: $P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \ldots \wedge P(x_n)$.

For example, in the above example of $\forall x\, P(x)$, if we knew that there were **only** 4 cars in our universe of discourse *(c1, c2, c3* and *c4)* then we could also translate the statement as: ***P(c1)*** $\wedge$***P(c2)*** $\wedge$***P(c3)*** $\wedge$***P(c4)***

The Existential Quantifier

The expression: $\exists x P(x)$, denotes the **existential quantification** of *P(x)*. Translated into the English language, the expression could also be understood as: "There exists an *x* such that *P(x)*" or "There is at least one *x* such that *P(x)*" $\exists$ is called the **existential quantifier**, and $\exists x$ means at least one object *x* in the universe. If this is followed by *P(x)* then the meaning is that *P(x)* is true for at least one object *x* of the universe. For example, "*Someone loves you*" could be transformed into the propositional form, $\exists x\, P(x)$, where:

- *P(x)* is the predicate meaning: ***x* loves you**,
- The universe of discourse contains (but is not limited to) all living creatures.

Existential Quantifier and **Connective OR**
If all the elements in the universe of discourse can be listed, then the existential quantification $\exists x P(x)$ is equivalent to the disjunction: $P(x_1) \vee P(x_2) \vee P(x_3) \vee \ldots \vee P(x_n)$.

For example, in the above example of $\exists x\, P(x)$, if we knew that there were **only** 5 living creatures in our universe of discourse (say: me, he, she, rex and fluff), then we could also write the statement as: ***P(me)*** $\vee$***P(he)*** $\vee$***P(she)*** $\vee$***P(rex)*** $\vee$***P(fluff)***

An appearance of a variable in a **wff** is said to be **bound** if either a specific value is assigned to it or it is quantified. If an appearance of a variable is not bound, it is called **free**. The extent of the application(effect) of a quantifier, called the **scope** of the quantifier, is indicated by

square brackets **[ ]**. If there are no square brackets, then the scope is understood to be the smallest **wff** following the quantification.
For example, in $\exists x\, P(x, y)$, the variable x is bound while y is free. In $\forall x\, [\exists y\, P(x, y) \vee Q(x, y)]$, x and the y in $P(x, y)$ are bound, while y in $Q(x, y)$ is free, because the scope of $\exists y$ is $P(x, y)$. The scope of $\forall x$ is $[\exists y\, P(x, y) \vee Q(x, y)]$.

How to read quantified formulas

When reading quantified formulas in English, **read them from left to right.** $\forall x$ can be read as "for every object x in the universe the following holds" and $\exists x$ can be read as "there erxists an object x in the universe which satisfies the following" or "for some object x in the universe the following holds". Those do not necessarily give us good English expressions. But they are where we can start. Get the correct reading first then polish your English without changing the truth values.

For example, let the universe be the set of airplanes and let $F(x, y)$ denote "x flies faster than y". Then
$\forall x \forall y\, F(x, y)$ can be translated initially as "For every airplane x the following holds: x is faster than every (any) airplane y". In simpler English it means "Every airplane is faster than every airplane (including itself !)".
$\forall x \exists y\, F(x, y)$ can be read initially as "For every airplane x the following holds: for some airplane y, x is faster than y". In simpler English it means "Every airplane is faster than some airplane".
$\exists x \forall y\, F(x, y)$ represents "There exist an airplane x which satisfies the following: (or such that) for every airplane y, x is faster than y". In simpler English it says "There is an airplane which is faster than every airplane" or "Some airplane is faster than every airplane".
$\exists x \exists y\, F(x, y)$ reads "For some airplane x there exists an airplane y such that x is faster than y", which means "Some airplane is faster than some airplane".

Order of Application of Quantifiers

When more than one variables are quantified in a wff such as $\exists y\, \forall x\, P(x, y)$, they are applied from the inside, that is, the one closest to the atomic formula is applied first. Thus $\exists y\, \forall x\, P(x, y)$ reads $\exists y\, [\forall x\, P(x, y)]$, and we say "there exists an y such that for every x, $P(x, y)$ holds" or "for some y, $P(x, y)$ holds for every x".

The positions of the same type of quantifiers can be switched without affecting the truth value as long as there are no quantifiers of the other type between the ones to be interchanged.
For example $\exists x\, \exists y\, \exists z\, P(x, y, z)$ is equivalent to $\exists y\, \exists x\, \exists z\, P(x, y, z)$, $\exists z\, \exists y\, \exists x\, P(x, y, z)$, etc. It is the same for the universal quantifier.

However, the positions of different types of quantifiers can **not** be switched.
For example $\forall x\, \exists y\, P(x, y)$ is **not** equivalent to $\exists y\, \forall x\, P(x, y)$. For let $P(x, y)$ represent $x < y$ for the set of numbers as the universe, for example. Then $\forall x\, \exists y\, P(x, y)$ reads "for every number x, there is a number y that is greater than x", which is true, while $\exists y\, \forall x\, P(x, y)$ reads "there is a number that is greater than every (any) number", which is not true.